

MODEL DEVELOPMENT & IMPLEMENTATION

In this section, an overview of the framework’s underlying constitutive equations is presented, then details regarding the custom adaptations and novel contributions developed in this project. The mathematical framework builds upon the CHASTE library [5] and an existing intestinal crypt specific package, *OrganoidCHASTE* [6]. Novel techniques are developed building upon Metzcar et al’s. [4] agent based fibre modelling framework, which is transferred to CHASTE, expanded to three dimensions and combined with a generalized Maxwell viscoelastic model of the ECM [7, 8]. Additionally, the work-energy lumen pressure formalisation from *OrganoidCHASTE* is expanded to function for 2D and 3D cell-centre models, and vertex 2D models.

2.1 Agent Based Modelling Methodologies

Agent Based Models (ABMs) treat each cell as a discrete entity, allowing the complex dynamics of an organoid to be modelled by decomposing the problem into discrete events such as cell-cell interactions, cell-membrane interactions and the cell cycle. While equation based continuum models could be considered to approximate the emergent behaviour of such discrete biological interactions, previous work has identified difficulty interoperating the complex hierarchy of biological interactions with the homogenisation assumption of continuum modelling techniques [9].

In the domain of agent based modelling for organoids, there are two main categories, on-lattice and off-lattice methods. On-lattice based methods use a regular mesh, where each element of the mesh can either represent a single cell (Cellular Automata), multiple cells (Lattice Gas) or together define the boundary of a cell (Cellular Pots). Off-lattice models offer better spatial resolution because cell position is not limited to the underlying mesh. Additionally, on-lattice models are bias towards the inherent unit directions of the grid, where for a square lattice, diagonal movement is penalised compared to moving to the directly adjacent elements. Therefore, in the following work only off-lattice cell position frameworks will be explored, specifically cell-centre based and vertex based methods. Note that an initial on-lattice ECM model is tested, however the unit direction bias was found to be problematic, leading to the final off-lattice ECM model (see section 2.3.3).

In cell-centre modelling each agent represents the centre of mass of the cell, then constitutive equations such as Hooke’s law can be used to model the forces and evolution of cells over time. Vertex models allocate a separate agent to each vertex of the cell, where each cell is associated with multiple vertices. To ensure results are generalisable (and not simply emergent properties of a singular modelling technique), two different off-lattice methodologies are used; Cell-centre and vertex based modelling (see sections 2.1.2 & 2.1.3 respectively).

2.1.1 The Cell Cycle

The cell cycle is fundamental to the development of intestinal crypts, where the continual generation of new cells acts as a conveyor belt, pushing cells out of the crypt to the villus tip where they are shed (apoptosis). Additionally, the differing mechanical properties between cell phenotypes has been shown to drive crypt fission [10], which is the self-organising process by which an embryonic intestinal organoid forms crypts. Specifically, Langlands et al. showed with in-vivo experimental results (and a computational model in CHASTE) that the lower stiffness and lower laminin binding affinity of adult intestinal stem cells were key drivers of this process [10]; where the stiffer Paneth cells (PC) have greater resistance to deformation and migration across the basement membrane. Further work by Almet et. al. [11] then demonstrated increased crypt fission with a higher proportion of PCs. However, these works were limited in their representation of the real biological cell cycle. To improve on this, Montes-Olivas et al.[2] introduced two additional cell types: a transit-amplifying (TA) and general differentiated enteroid cell (EC). Where TA cells are proliferative, multipotent cells and enteroid cells represent all other terminally differentiated cell types (such as enteroendocrine, enterocyte, goblet or Tuft cells), see figure 2.1 for a visualisation of this cycle.

Because of its greater repertoire of cell types, and as the CHASTE implementation is openly available, Olivass et al’s adapted cell-cycle is used for all model variations presented in this project. A limitation of this model is its exclusion of signalling pathways, which, while potentially limiting the accuracy of the model, allow the scope to be focused on biomechanics over molecular biology.

Contact Inhibition is the process by which epithelial cells downregulate proliferation after saturating the local growth medium (often described as confluence). The Wnt and Hippo pathways have been shown to drive contact inhibition [12, 13]. However, as such signalling pathways are not simulated explicitly in this project, a purely mechanical contact inhibition mechanism is implemented. Specifically, cells can only divide if they are not compressed beneath a minimum volume or area, for 3D or 2D models respectively. The rule for contact inhibition used in this work is, if $\frac{V_{curr}}{V_{eq}} < f_Q$ then cell division is prevented. Where V_{curr} is a cell’s current cell volume (or area), V_{eq} is a cell’s target equilibrium volume (or area) and f_Q is the quiescent fraction (which acts as the threshold). The parameters of the cell cycle and contact inhibition mechanism used for all models are detailed in table 2. This mechanism is taken from CHASTE’s default *ContactInhibitionCellCycleModel* class.

Cell death (apoptosis) is not modelled in this project which is an obvious simplification. In

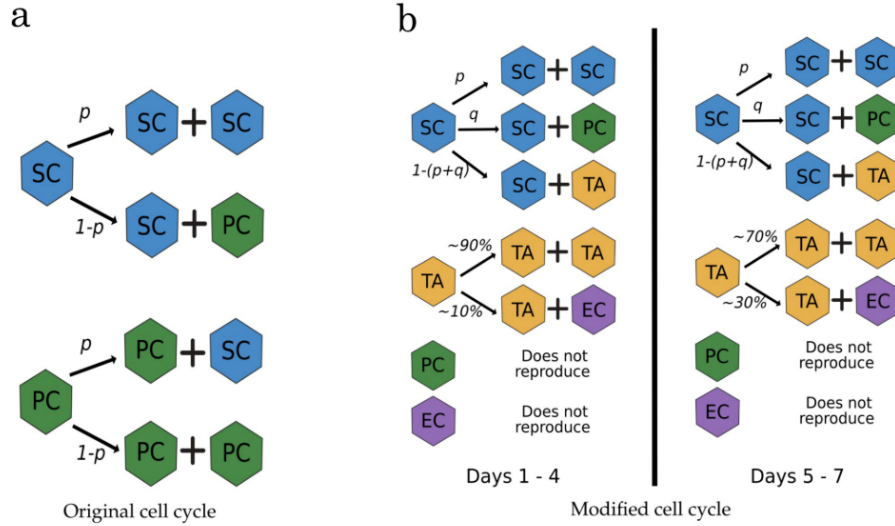


Fig 2. Cell cycle diagrams. (a) Original cell-cycle model used by Langlands et al. and Almet et al. [25, 30], (b) 'proposed model' modification that requires a production probability for each cell type derived from stem cells. Both models maintain the stochasticity of the system, as the selection of cell type is dependent on a random number generated at each time-step. Cell types: stem cell (SC); Paneth cell (PC); transit amplifying cell (TA); general differentiated enteroid cell (EC).

<https://doi.org/10.1371/journal.pcbi.1011386.g002>

Figure 2.1: Reproduced from Olivas et al., [2], Fig 2. This figure shows the both the biologically inconsistent original cell cycle (which can produce stem cells from Paneth cells) and the modified cell cycle which introduces additional cell types and biologically accurate transitions.

real intestinal tissue cell apoptosis can be triggered by a variety of signalling pathways and mechanisms [14]. In the intestinal crypt niche, this process is dominated by cell extrusion from the monolayer at the villus tip. The omission of the cell death process is acceptable as long as the simulation length remains lower than the average differentiated cell life time of 4-5 days [15]. **I am looking to implement a trigger for “anoikis” once a cell is no longer in contact with the ECM, see seminal paper [16]. This paragraph will be adapted accordingly.**

2.1.2 Cell-centre Based Models

The key force equations which control mechanical intercellular interaction in CHASTE's cell-centre models are derived from the linear spring equation [17] as below,

$$(2.1) \quad \mathbf{F}_i = \sum_{j \in \mathcal{N}(i)} \mu(|\mathbf{r}_{i,j}| - l_{i,j}^0) \hat{\mathbf{r}}_{i,j}$$

Where the total force vector ($\mathbf{F}_i$) acting on a cell i at each time step, is equal to the sum of all forces from the springs of the set of all neighbouring cells $\mathcal{N}$. Where the set of neighbouring cells are those which are within a threshold distance (see interactionCutoff2d of table 1). The individual spring forces are modelled as the spring constant μ_{ij} (dependent on the edge's type between $\langle i, j \rangle$) multiplied by the difference between the actual intercellular distance ($|\mathbf{r}_{i,j}|$) and equilibrium length $l_{i,j}^0$ (Hooke's Law). Finally, this scalar force is multiplied by the unit intercellular distance vector ($\hat{\mathbf{r}}_{i,j}$). In summary, assuming linear damped springs between cells and the resulting change in cell position as,

$$(2.2) \quad \Delta \mathbf{p}_i = \frac{\Delta t}{\eta} \mathbf{F}_i$$

Where $\Delta \mathbf{p}_i$ is the change in position of the centre of cell i , η the damping constant representing the friction of the ECM, and Δt is the simulation time step. While this system is simple, its use of the linear spring equation makes it highly interpretable, allowing constants to be determined from experimentally observed physical properties of the cell micro-environment [18].

However, use of this system of equations relies on a number of assumptions regarding the dynamics of cells in the epithelium. Primarily that cells tend to establish an equilibrium distance (equilibrium length l_{ij}^0) to neighbouring cells through damped spring like movement. There are two notable cell phases when the equilibrium distance is controlled, cell birth (Mitosis) and cell death (Apoptosis). During mitosis the equilibrium length starts at a small lambda value then increases linearly over time (up to the steady state value of 1 cell length). In a similar fashion, during Apoptosis the equilibrium length reduces linearly until the cell is removed.

2.1.3 Vertex Based Models

Vertex based models are typically constructed using surface energy minimisation, where compared to cell-centre based models, the boundary of each cell is well defined. The Nagai-Honda formalisation was the seminal work for vertex based modelling of epithelial cells using energy minimisation, inspired by the formation of soap foams [19]. This formalisation equates the total potential energy of the epithelial monolayer to

$$(2.3) \quad U = \sum_{\alpha} \left[\lambda_d (A_{\alpha} - A_{\alpha}^0)^2 + \lambda_m (L_{\alpha} - L_{\alpha}^0)^2 + \sum_{\langle i,j \rangle \in \alpha} \gamma_{ij} \ell_{ij} \right]$$

where the sum is over all cells α . The first term is the area elastic energy A_{α} , resisting deformation from the target area A_0 with stiffness λ_d . The second term penalises difference between the actual L_{α} and target cell perimeter $L_{\alpha}^0 = 2\sqrt{\pi A_{\alpha}^0}$ (assuming circular cell faces). This term also has a scalar parameter λ_m which reflects the strength of the actomyosin ‘belt’ complex around the cell perimeter. The third term assigns a line tension γ_{ij} to each edge $\langle i,j \rangle$ of length ℓ_{ij} , modelling the effect of intercellular adhesion. As detailed in appendix table 1, γ_{ij} is either set to $\gamma_{cell-cell}$ for intercellular edges, or $\gamma_{boundary}$ for edges which are adjacent to the ECM or lumen. This facilitates the application of differential adhesion as described in section 2.1.3. The force on each cell ($\mathbf{F}_i$) is then computed as the negative gradient of the energy functional from equation 2.3 as $\mathbf{F}_i = -\nabla_i U$. This force is then used to evolve the position of each vertex with the over-damped equation of motion and Euler stepping, as used in the cell-centre model (see equation 2.2). Since Nagai et al.’s original formulation, extended energy functionals such as those proposed by Fletcher et al. [3] have become widely adopted.

Topological Transitions

In vertex models, as new cells are born or die and the monolayer deforms, the shape of faces which define the boundaries of cells also deform. If this causes a cell face to shrink below a certain threshold, a so called topological transition is used to rearrange the vertex connectivity [3]. Without such mechanisms, the mesh topology would be static, preventing the tissue from undergoing realistic

large-scale deformations. Two principal transition types are implemented: T1 (neighbour exchange) and T2 (cell removal). Other transitions have been modelled such as T3 (vertex/edge merging), however these transitions increased the occurrence of numerical instability, requiring reduced time step size and thus increased computational demand. Therefore more complex transition were omitted, this did not cause any noticeable impact of large-scale tissue dynamics, however specific research on the choice of possible topological transitions would be insightful.

T1 transitions, referred to as a neighbour exchange or intercalation, these transitions occur when a shared edge between two cells shrinks below a critical threshold length ℓ_{T1} and is then removed. When the edge length ℓ_{ij} satisfies $\ell_{ij} < \ell_{T1}$, it is removed and replaced by a new edge, oriented perpendicular to the original (see figure 2.2). The length of the newly created edge is set to a small but finite value (typically $1.5 \ell_{T1}$) to prevent immediate retriggering of the transition. Cell intercalation, which is enabled with this transition, is key to morphological changes in epithelial sheets [20].

T2 Transition, (cell removal) is triggered when the area A_α of a cell α falls below a critical threshold area A_{T2} , $A_\alpha < A_{T2}$. The T2 transition removes the cell from the tessellation: all vertices belonging exclusively to cell α are deleted and replaced by a single vertex located at the centre of the removed cell. The surrounding cells then share this new vertex and their edges are reconnected, closing the gap left by the removed element. T2 transitions model biological processes such as apoptotic cell death, where a dying cell is expelled from the epithelial sheet [3].

Differential adhesion

Variable adhesion energy between different cell types, known as differential adhesion, is fundamental to the rearrangement of cells during morphogenesis [21]. To incorporate this into the model, spring stiffness is altered in the cell-centre based models and intercellular adhesion energy is altered in the vertex based models, depending on cell type. The values of the spring stiffness (μ_{ij}) and differential adhesion parameter (γ_{ij}) for each cell-cell pair type are parametrised in the final application as detailed in figure 3.

2.2 Lumen Pressure Force Model

Intestinal organoids have a central, fluid-filled cavity known as the lumen. The apical surface of the epithelium bounds the lumen, which is assumed to apply a hydrostatic pressure on the surrounding tissue. This radial expansion force generated by the lumen is essential for organoid morphogenesis, driving the initial spherical expansion that precedes crypt budding events. The hydrostatic assumption is based on the long time-scale (hours or days) of the morphological changes compared to the predominantly aqueous (low viscosity) lumen. This assumption also allows integration into the quasi-static modelling paradigm used in the vertex and cell-centre models thus far.

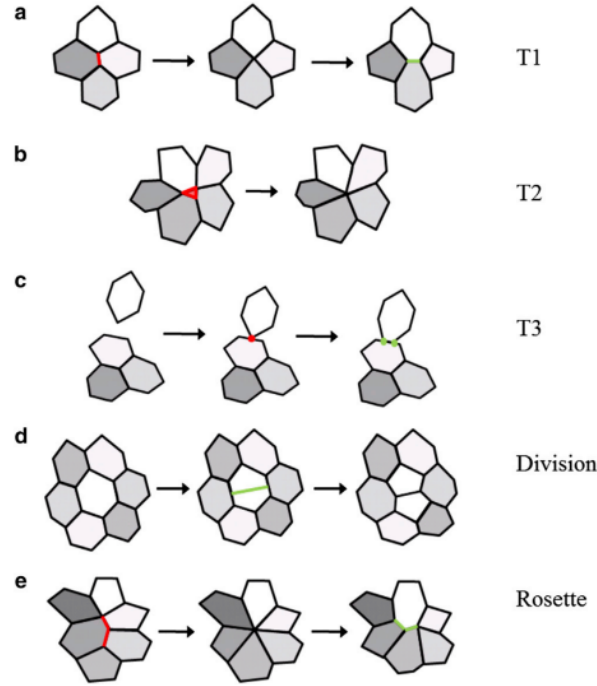


Figure 2.2: Schematic of topological transitions in a 2D vertex model reproduced from [3]. **(a)** T1 transition: when the shared edge ℓ_{ij} between cells A and B contracts below the threshold ℓ_{T1} , it is replaced by a perpendicular edge connecting cells C and D, exchanging neighbour relationships. **(b)** T2 transition: when the area of a cell falls below the threshold A_{T2} , the cell is removed from the tessellation and its vertices are collapsed to a single point, with surrounding cells reconnected.

From a continuum mechanics perspective, the lumen can be treated as an incompressible or nearly-incompressible fluid exerting a uniform pressure P normal to the apical epithelial surface. This formalisation follows the classical pressure–volume work relation, where the work done by the lumen on the epithelium during infinitesimal expansion is $\delta W = P dV$.

The cell-centre based models pose a problem, because unlike the vertex models, the surface area of each cell is not well defined. Simpler approaches could have been used to apply a constant outward pressure for the cell-centre based simulations. However, by using the work-energy frame, and thus keeping a consistent method across all models, the results will be more directly comparable.

2.2.1 Mathematical formalisation

From the work-energy formula,

$$(2.4) \quad \delta W = P dV$$

where P is the constant lumen pressure, W is the work exerted by that pressure on the cell and V is the volume (3D) or area (2D) of the lumen cavity. Given the zero-inertia (overdamped) motion assumption used for all models (see equation 2.2), it follows that the force on each cell (F_i) is

equivalent to

$$(2.5) \quad \mathbf{F}_i = P \frac{\partial V}{\partial \mathbf{r}_i} = P \nabla_i V$$

where ∇_i denotes the gradient with respect to the position coordinates of cell i . This represents the instantaneous sensitivity of the lumen volume to a displacement of that specific node and relies on the principle of virtual work.

Lumen Volume Gradient Calculation

For each model type there is a distinct method used to find the lumen volume gradient w.r.t the cell position ($\frac{\partial V}{\partial \mathbf{r}_i}$).

3D vertex

The existing implementation of the hydrostatic lumen pressure in the OrganoidCHASTE library [6] was used as the reference for the following lumen volume gradient formalisations. The lumen volume is computed using the signed-tetrahedra method applied to the apical surface of the monolayer. Each apical face is a polygon with n_f vertices, which is triangulated into n_f sub-triangles by connecting each edge to the face centroid $\mathbf{c}_f$. Each sub-triangle forms a signed tetrahedron with the coordinate origin, and the total lumen volume is:

$$(2.6) \quad V = \frac{1}{6} \left| \sum_f \sum_{j=0}^{n_f-1} \mathbf{r}_j \cdot (\mathbf{c}_f \times \mathbf{r}_{j+1}) \right|$$

To obtain the pressure force on each vertex, the gradient of the lumen volume w.r.t each node position $\mathbf{r}_i$ is required. Since the face centroid $\mathbf{c}_f = \frac{1}{n_f} \sum_k \mathbf{r}_k$ depends on $\mathbf{r}_i$, the chain rule must be applied when differentiating through $\mathbf{c}_f$. For a face f containing node i with cyclic neighbours $\mathbf{r}_{i-1}$ and $\mathbf{r}_{i+1}$:

$$(2.7) \quad \nabla_i V = \frac{1}{6} \sum_{f \ni i} \left[(\mathbf{c}_f \times \mathbf{r}_{i+1}) + (\mathbf{r}_{i-1} \times \mathbf{c}_f) + \frac{1}{n_f} \sum_{j=0}^{n_f-1} (\mathbf{r}_{j+1} \times \mathbf{r}_j) \right]$$

The first two terms come from differentiating the sub-triangles adjacent to node i , while the third corrects for the dependence of $\mathbf{c}_f$ on $\mathbf{r}_i$. The pressure force applied to each apical node is then $\mathbf{F}_i = P \nabla_i V$, directed along the local outward normal of the apical surface.

2D cell-centre

With the understanding of the implementation in OrganoidCHASTE, the 2D cell-centre case is considered. All cell centres are sorted counter-clockwise by angle around the lumen centroid to form a polygon. The signed area is given by the shoelace formula,

$$(2.8) \quad A = \frac{1}{2} \sum_{i=0}^{N-1} (x_i y_{i+1} - x_{i+1} y_i)$$

Where x and y are the Cartesian coordinates of i^{th} and $i^{\text{th}} + 1$ (counter-clockwise adjacent) cell centre. Differentiating with respect to the position of cell i results in:

$$(2.9) \quad \nabla_i A = \begin{pmatrix} \frac{1}{2}(y_{i+1} - y_{i-1}) \\ \frac{1}{2}(x_{i-1} - x_{i+1}) \end{pmatrix}$$

giving the force on each cell as,

$$(2.10) \quad \mathbf{F}_i = P \nabla_i A = \frac{P}{2} \begin{pmatrix} y_{i+1} - y_{i-1} \\ x_{i-1} - x_{i+1} \end{pmatrix}.$$

2D vertex

In vertex models the agents are now mesh vertex positions, not cell centres. Therefore, to apply the same shoelace based volume differential, the centroids of the apical cells are used to form a polygon, analogous to the cell-centre case. Applying the shoelace formula as before yields a per-cell force $\mathbf{F}_i^{\text{cell}}$. This force is then distributed equally to the n_i nodes of cell i :

$$(2.11) \quad \mathbf{F}_{i,k}^{\text{node}} = \frac{\mathbf{F}_i^{\text{cell}}}{n_i} = \frac{P}{2n_i} \begin{pmatrix} y_{i+1} - y_{i-1} \\ x_{i-1} - x_{i+1} \end{pmatrix}$$

Splitting the force equally between the vertices of each cell is a simplification, instead the volume differential could be calculated for each vertex individually. However, given that there are multiple vertices for each cell (4 on average), this would quadruple the number of operations for the lumen pressure force. Also, given the proximity of cell vertices compared the epithelium as a whole, the difference in volume differential should be small. **I realised this was a simplification while writing this, I can very easily convert to using the vertex based differential**

3D cell-centre

Assuming a spherical arrangement of N cells, each cell i at distance $r_i = |\mathbf{r}_i - \mathbf{c}|$ from the lumen centroid $\mathbf{c}$, the radial component of the volume gradient is $\frac{\partial V}{\partial r_i} = \frac{4\pi r_i^2}{N}$, where the change in lumen volume is assumed to be shared equally between each cell. The outward unit normal is the radial direction:

$$(2.12) \quad \hat{\mathbf{n}}_i = \frac{\mathbf{r}_i - \mathbf{c}}{|\mathbf{r}_i - \mathbf{c}|}$$

Thus the force vector on the i^{th} cell is

$$(2.13) \quad \mathbf{F}_i = P \nabla_i V \approx P \frac{4\pi r_i^2}{N} \hat{\mathbf{n}}_i$$

The spherical cell distribution is clearly violated in the presence of crypts. However, technical development was focused on the 2D cell-centre and 2D vertex models to ensure a full set of baseline results. A possible improvement would be to form a mesh from the cell-centre positions and use the mesh volume formalisation as is used by the vertex 3D method. The only challenge would then be to ensure the mesh was valid, where any overlapping cells could cause an error in volume calculation. The inaccuracy of this volume differential is expected to enforce greater sphericity and fewer crypt budding events, this hypothesis is evaluated in section 4. **This section reference will be updated once it has been complete**

2.3 Extra-cellular Matrix Modelling

A key limitation to many computational models of organoids is the simplified interaction with the extracellular matrix (ECM). This interaction has previously been incorporated using simple bending resistance forces such as by Dunn et al., [22]. This does not include the plethora of mechanisms which mediate the cell-ECM interaction, including but not limited to: ECM degradation by Matrix Metalloproteinases (MMPs) [23], ECM contact mediated cell death (Anoikis) [24], collagen fibre orientation [4] and molecular clutch dynamics [25]. Critically, it has been established that the make up of the ECM and resultant ECM-Cell interactions have instrumental effects on the morphological dynamics observed in organoid growth [26]. This project focuses on the macro-scale mechanical components of the cell-ECM interaction (in the micrometer (μm) range), omitting many complexities of molecular biology to homogeneity assumptions (linear springs, hydrostatic pressure etc.)

Initially, taking inspiration from Metzcar et al's [4] agent based modelling of the ECM, a parallel implementation was made in CHASTE. To verify the implementation, Painter et al's cell invasion scenario [27] is recreated, as was done in Metzcar et al's work. While this on-lattice approach was capable of reproducing known cell invasion dynamics, it posed challenges when organoid growth is considered. For example, as the organoid expands, it exerts an outward radial pressure on the ECM. The on-lattice based method failed to replicate this as the organoid could move over the lattice instead of needing to 'displace' the ECM. Therefore a second, off-lattice, agent based ECM model was introduced into CHASTE (See section 2.3.3). A similar approach has been used by Carrasco-Mantis et al. [28].

2.3.1 Initial ECM Fibre Modelling

anisotropy to go in glossary of terms? A new CHASTE class is used to replicate Metzcar et al's work [4]. As visually represented in figure 2.3, this framework directly sets the velocity of the cell depending on the local anisotropy, direction, and density of fibres. As cells move they also remodel the local ECM fibre direction. The corresponding equations are,

$$(2.14) \quad \mathbf{v}_{\text{loc}} = s_{\text{cell}} \mathbf{d}_{\text{actual}}$$

$$(2.15) \quad \mathbf{d}_{\text{actual}} = (1 - \gamma) C_1 \mathbf{d}_{\perp} + C_2 \mathbf{f}$$

$$(2.16) \quad \gamma = a(\mathbf{x}_i^*) \cdot s_i$$

$$(2.17) \quad s_{\text{cell}} = \begin{cases} 0 & \text{if } \rho \leq \rho_l \\ S_{\text{max}} \frac{\rho - \rho_l}{\rho_{\text{ideal}} - \rho_l} & \text{if } \rho_l < \rho \leq \rho_{\text{ideal}} \\ S_{\text{max}} \frac{\rho - \rho_h}{\rho_{\text{ideal}} - \rho_h} & \text{if } \rho_{\text{ideal}} < \rho < \rho_h \\ 0 & \text{if } \rho \geq \rho_h \end{cases}$$

Table 2.1: Summary of variables and parameters for the ECM fibre interaction model (equations 2.14–2.17).

Symbol	Description	Range / Units
<i>Cell motion variables</i>		
$\mathbf{v}_{\text{loc}}$	Locomotive velocity vector	$\mu\text{m min}^{-1}$
s_{cell}	Cell speed (determined by local ECM density)	$[0, S_{\text{max}}]$
$\mathbf{d}_{\text{actual}}$	Actual motility direction (unit vector)	–
<i>ECM structure variables</i>		
ρ	Local ECM density (fibre volume fraction)	$[0, 1]$
$\mathbf{f}$	Average local ECM fibre orientation (unit vector)	–
$a(\mathbf{x}_i^*)$	Local anisotropy of fibre alignment	$[0, 1]$
$\mathbf{d}_{\perp}$	Unit vector perpendicular to $\mathbf{f}$ in the plane of the cell's preferred direction	–
<i>Coupling parameters</i>		
γ	Effective ECM influence (contact guidance strength)	$[0, 1]$
C_1, C_2	Cell direction preference parameters	–
<i>Constants and thresholds</i>		
s_i	Cell-specific ECM sensitivity	–
S_{max}	Maximum cell migration speed	$\mu\text{m min}^{-1}$
ρ_{ideal}	Ideal density (at which $s_{\text{cell}} = S_{\text{max}}$)	$[0, 1]$
ρ_l	Lower density threshold ($s_{\text{cell}} = 0$ below)	$[0, 1]$
ρ_h	Upper density threshold ($s_{\text{cell}} = 0$ above)	$[0, 1]$

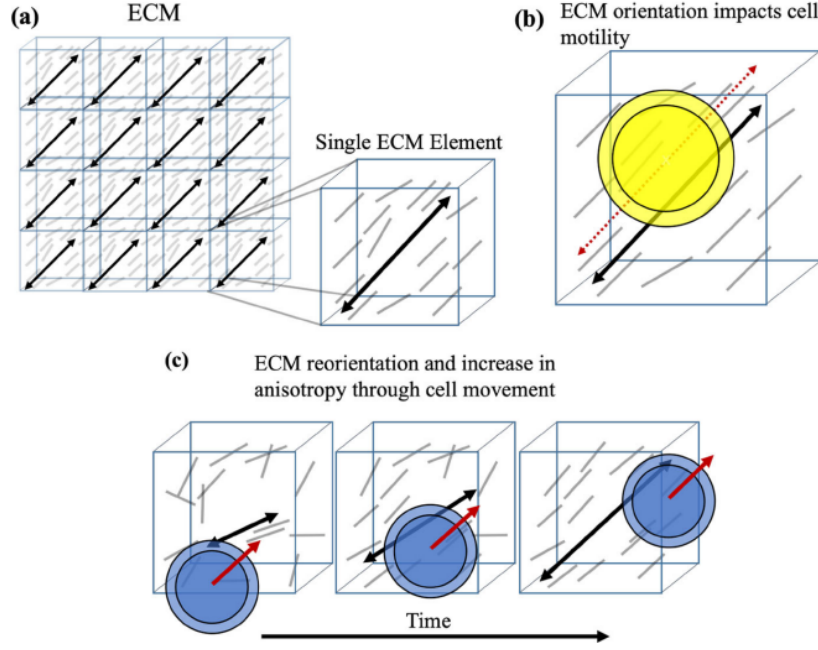


Figure 2.3: Reproduced from Metzcar et al [4]. **a)** Shows the gridded arrangement of ECM elements where instead of modelling individual fibres, the mean fibre direction is used. In this diagram the smaller gray arrows depict the actual ECM fibres which are aggregated to the mean direction for each ECM element's fibre direction. **b)** Depicts the change in cell direction to align to the local ECM element's direction (as per equation 2.17). **c)** Shows remodelling of the ECM as the cell exerts traction forces to the local fibres (these equations are detailed in the technical supplement of Metzcar et al's paper [4])

ECM Degradation

Cells remodel the surrounding ECM environment via degradation and deposition. In this initial CHASTE implementation the ECM is degraded linearly over time towards 0 ($d\rho/dt = -k_{deg}$), such that each cell degrades neighbouring ECM elements at this rate (accumulating with multiple cells). Metzcar et al's model used a cell phenotype specific target density and exponential decay. At this point during development no discrete cell phenotypes were allocated so this approach was not applicable.

Validation

To validate the new CHASTE implementation was correct, and reproduces the same behaviour as Metzcar et al's, PhysiCell model [29], a cell invasion study was performed. Three different ECM configurations are used, where the ECM fibres are oriented randomly, parallel, perpendicular or mixed (w.r.t the lower boundary of the domain). The mixed configuration sets the two outer thirds of the ECM as perpendicular and the centre as parallel.

From the results in Figure 2.4, the new implementation is confirmed to display similar qualitative behaviour as observed in Metzcar et al's cell invasion replication [4]. This in turn aligns with experimentally observed results from Painter et al [27]. With greatest cell invasion observed

with the perpendicular and mixed matrix fibre configuration, minimum invasion with parallel configuration, and random configuration in-between.

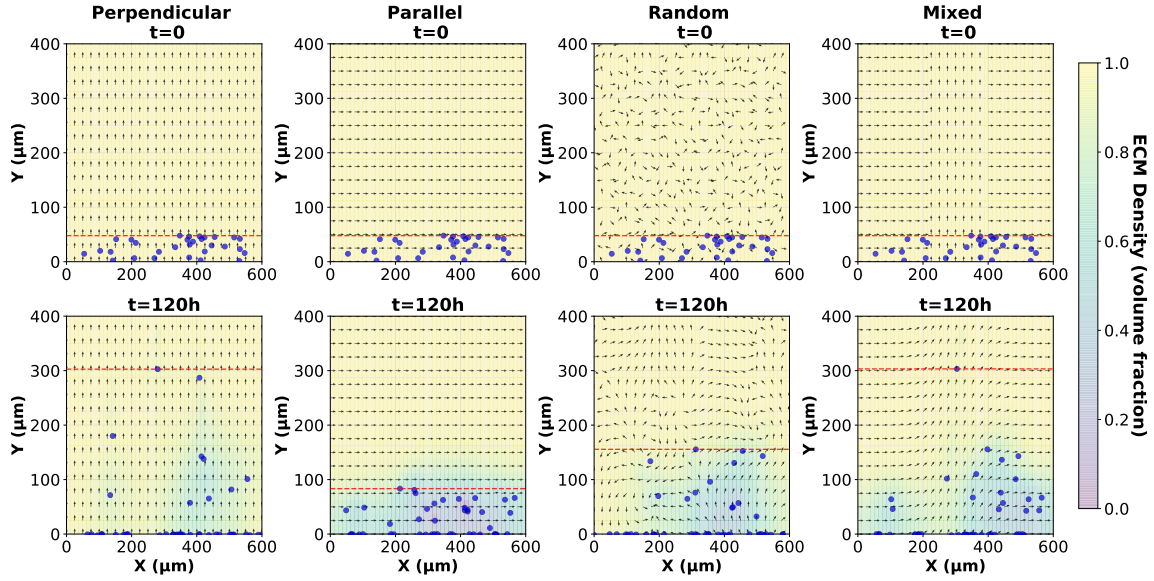


Figure 2.4: Each plot shows the initial and final time steps of a 120 hour (5 day) simulation under 4 different ECM configurations. The blue dots represent cell centres, and arrows show the local direction of the ECM. The colour of the ECM corresponds to the ECM density, measured in this model as the protein fibre volume fraction. The red dotted line marks the furthest cell invasion for each configuration.

2.3.2 On-Lattice Monolayer Scale Model

After replicating the expected cell-ECM interaction of the agent based fibre model, the next step was to begin modelling the growth of an intestinal organoid rather than individual cells with no inter-cell interactions. The first change, made to adapt the ECM interaction for the full organoid scale simulation, was to transition from a speed based ECM interaction to force based interaction. This was key because the cell invasion logic directly set the cell speed based with solely the ECM properties (see equation 2.14), while in the full organoid simulation the contributions from a multitude of additional forces needs to be combined.

Taking inspiration from the quasi-static linear spring approach (as in section 2.1.2), each element of the ECM exerts a force on any cells with in a distance threshold of that given ECM element. To adapt this force to the ECM-Cell interaction, the force of each linear spring element from the ECM is modulated by the density, relative fibre direction and anisotropy. The updated force contribution then becomes

$$(2.18) \quad \mathbf{F}_i^{\text{ECM}} = \sum_{j \in \mathcal{N}(i)} \mu_{\text{ECM}} (|\mathbf{r}_{ij}| - l_0) \rho_j \Lambda_{ij} \hat{\mathbf{r}}_{ij}$$

where $\mathbf{F}_i^{\text{ECM}}$ is the total ECM force on cell i , and $\mathcal{N}(i)$ is the set of ECM elements within the interaction cutoff of that cell. The spring constant is μ_{ECM} , l_0 is the spring rest length, and $\mathbf{r}_{ij} = \mathbf{x}_j^{\text{ECM}} - \mathbf{x}_i$ is the displacement vector from cell i to ECM element j , with unit direction

$\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/\|\mathbf{r}_{ij}\|$ and the local ECM density is $\rho_j \in [0, 1]$. Λ_{ij} is the alignment multiplier, which increases spring strength when cell motion is aligned with local fibres and decreases it when misaligned, given by the following

$$(2.19) \quad \Lambda_{ij} = \left(1 - \alpha_j^{\text{eff}}\right) + \alpha_j^{\text{eff}} |\hat{\mathbf{m}}_i \cdot \hat{\mathbf{f}}_j|$$

where the local anisotropy is $\alpha_j \in [0, 1]$, and α_j^{eff} is the density-weighted anisotropy which is equal to α_j multiplied by ρ_j . The unit vector $\hat{\mathbf{m}}_i$ denotes the last movement direction of cell i , while $\hat{\mathbf{f}}_j$ denotes the local ECM fibre direction at element j .

This formalisation successfully adapts Metzcar et al.,’s speed based ECM interaction to a force contribution. Note that it omits the lower, upper and ideal densities (ρ_l , ρ_h & ρ_{ideal}) in favour of a continuous force profile. The ideal density modelling approach was based on the observation that a cell’s migration is impeded if the ECM density is too low (where the cell lacks adhesion), or when it is too high (obstructing movement) [30, 31]. In this force contribution-based approach, this behaviour arises implicitly. For example, if a cell’s neighbours are exclusively low density ECM elements, the force from the ECM elements will be low, and so will the ECM’s contribution to cell migration speed. Alternatively, when an ECM element is surrounded by high density ECM elements, it must overcome the strong adhesive spring before moving away, thus limiting migration. Instead of an ideal density, this model has an ideal density distribution, with low density ECM elements behind a cell’s migration path (to limit the adhesive force pulling the cell back) and high density ECM elements in front (pulling the cell forward). A notable difference in cell behaviour during this ideal density distribution (compared to the speed-based model) is that while allowing the cell to initially migrate rapidly, the cell’s movement will be limited once the high density ECM lattice element is behind the cell. I think I should add a diagram to more effectively explain the paragraph

Lattice Structure and Initial Conditions

A uniform hexagonal mesh defines the elements of the ECM with default spacing of $25\mu\text{m}$. The hexagonal lattice was preferred because it has the greatest number of faces for a regular space filling polygon. This reduces grid-alignment artefacts without requiring storage of an explicit adjacency table. For example, a centroid Voroni tessellation (CVT) would be an improvement on the hexagonal mesh with regards to grid-alignment artefacts but would suffer from increased memory latency because of the explicit adjacency matrix lookup required.

To represent the absence of ECM inside the organoid lumen, a circle with the same radius as the initial cell distribution (described in section 2.4) is used to set all interior ECM element’s density to 0. This causes the resultant force from these elements to be eliminated (as per equation 2.18). However, this method of recapitulating the lumen does not reflect the viscoelastic flow of hydrogels. Specifically, as the ECM is degraded the organoid migrates across the ECM, leaving behind an area of low density ECM (FIGURE DEMONSTRATING). In the current model, the ECM elements which are set to zero density to represent the lumen, and those which are degraded by cell interactions, no longer exert any force on the cell population. In contrast, a real viscoelastic

hydrogel would display fluid like flow back into such areas of low density. This discontinuity with real hydrogel properties and the inability to remove grid alignment artifacts motived the next iteration of the model in the following section (2.3.3).

2.3.3 Ghost Node Off-Lattice Model

The following section describes the final ECM model, an off-lattice viscoelastic model which aims to address the limitations of the initial on-lattice approach (of section 2.3.2). These issues were the lack of real viscoelastic behaviour, and persistent grid alignment artifacts. By associating a dynamic position to each ECM element, common viscoelastic phenomenon such as stress-relaxation and a time-dependent stress-strain curve could be accurately recreated ([reference a section of the results here?](#)). Additional to each ECM ghost node's position, the same ECM properties as in the on-lattice method are maintained (anisotropy α_j , density ρ_j and fibre direction $\mathbf{f}_j$). These are used to reflect the same physical properties as in Metzcar et al's simple agent based modelling framework [4], except now with off-lattice elements. First the linear cell-ECM interaction is described, then the viscoelastic ECM-ECM interaction and finally the ECM degradation and Fibre remodelling components.

Cell-Ghost Node Spring Force

Again adapting the spring force framework used with the cell-centre based intercellular interactions (see equations 2.1 & 2.18), the force exerted on cell i by neighbouring ghost nodes is given by

$$(2.20) \quad \mathbf{F}_i^{\text{cg}} = \sum_{j \in \mathcal{N}_g(i)} f^{\text{cg}}(d_{ij}, \rho_j) \Lambda_{ij}^g \hat{\mathbf{r}}_{ij},$$

where for each cell i , we define $\mathcal{N}_g(i)$ as the set of ghost nodes within the interaction cut-off $r_{\text{cut}}^{\text{cg}}$. d_{ij} is the cell-ghost separation, ρ_j is the ECM density for ghost node element j , $\hat{\mathbf{r}}_{ij}$ is the unit displacement direction, and Λ_{ij}^g is the anisotropic modulation factor (as detailed in equation 2.22). The scalar force magnitude f^{cg} is a piecewise function depending on whether the spring is in compression or extension (relative to the rest length l_0^{cg}). This function is key to prevent numerical errors from large displacements and to prevent physically unrealistic migration of the ECM. Numerical errors are mitigated at large deformations by the exponential decay of adhesive force (extension), and cells are prevented from migrating passed ECM nodes by the natural log term, such that the compression force tends to infinity as the distance tends to 0. Specifically, given the displacement from equilibrium $\delta_{ij} = d_{ij} - l_0^{\text{cg}}$, the function is

$$(2.21) \quad f^{\text{cg}}(d_{ij}, \rho_j) = \begin{cases} \mu_{\text{cg}} \rho_j^{\text{cap}} l_0^{\text{cg}} \ln \left(1 + \frac{\delta_{ij}}{l_0^{\text{cg}}} \right), & \delta_{ij} \leq 0 \text{ (compression)}, \\ \mu_{\text{cg}} \rho_j \delta_{ij} \exp \left(-\alpha \frac{\delta_{ij}}{l_0^{\text{cg}}} \right), & \delta_{ij} > 0 \text{ (extension)}. \end{cases}$$

Here μ_{cg} is the cell-ghost spring stiffness and $\alpha = 5$ sets the decay of the attractive force at large separations. In compression ρ_j^{cap} is used, which is simply ρ_j limited between $[0.1, 1]$. This prevents

low density ECM nodes passing through cells, where a ρ_j^{cap} value < 0.1 caused a vanishing force contribution (reference a figure to show this?). In extension, the attractive force is linearly weighted by ρ_j , so that degraded ECM exerts weaker adhesion on cells. Limiting the density term was not required in this case because as cells approach the cut-off distance r_{cut}^{cg} , the adhesive force of 0 would not cause any unrealistic behaviour. In comparison, the compressive force should not be entirely eliminated because the high aqueous fraction of hydrogels (CITATION) means there should be an incompressible component. This piecewise function was adapted from CHASTE's GeneralisedLinearSpringForce for cell-cell interactions, implementing non-linear components at extension and compression extremes for the same reasons (numerical stability and to better reflect incompressibility in close proximity). Note that each ghost node j receives the equal and opposite reaction force ($-f^{cg}\Lambda_{ij}^g\hat{\mathbf{r}}_{ij}$), from Newton's third law. This was not applicable in the on-lattice model because the ECM elements were static. Without this addition cells would be repelled from ECM elements but expanding cells would not repel the surrounding ECM. Likewise, under spring extension, the cells would be adhered to the ECM elements but ECM elements would remain static.

Anisotropic Modulation. The anisotropy of a given ECM element is intended to reflect the local alignment of ECM fibres. Where a low value (< 1) reflects low coherence between fibre directions and a high value (≈ 1) reflects full directional alignment. Considering the fibre properties in terms of a population, rather than individual collagen proteins, abstracts the vastly more complex molecular dynamics apparent at the smaller length scale citation?. Instead, this model only considers the emergent properties of the population of fibres at approximately the cell length scale (10–20 μm). As long as the assumptions underlying the expected force contributions from the population of collagen fibres holds, then the changes in morphology are expected to align with in-vivo behaviour. This could be expanded upon with relevant literature

To incorporate anisotropy into the spring force, an anisotropy modulation factor is devised. Given by

$$(2.22) \quad \Lambda_{ij}^g = 1 - \beta a_j |\hat{\mathbf{r}}_i \cdot \hat{\mathbf{f}}_j|,$$

where $\beta \in [0, 1]$ is an anisotropy strength parameter, determining the extent of the force reduction from anisotropy. $\mathbf{r}_i$ is the unit direction of motion of cell i , and $\mathbf{f}_j$ is the fibre direction of node j .

Initially β is set to 0.5, to reflect that adhesive and repulsive forces are still expected when cells are moving in full alignment with the ECM fibre direction, due to cross-linking and van der waals forces (more literature required). Therefore β should always be greater than 0. However, determining the ideal relationship between anisotropy and force reduction is non-trivial. The dot product measures if the direction of cell migration is aligned with fibre direction. Such that ($|\hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{f}}_j| \approx 1$) when aligned, resulting in a minimum value of Λ_{ij}^g and thus a minium of f^{eg} (all else equal). In contrast, when ($|\hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{f}}_j| \approx 0$), Λ_{ij}^g and the cell-ECM forces are at a maximum.

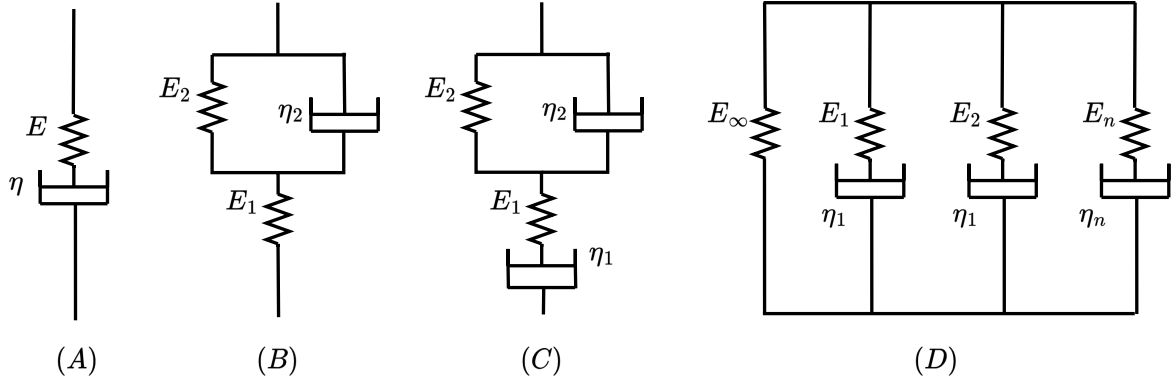


Figure 2.5: Summary of common lumped parameter model configurations for viscoelastic materials. E represents the elastic modulus of the spring components and η is the viscoelastic damping coefficient. A) Maxwell unit (or arm) consisting of a single dashpot and linear spring. B) A Kelvin-Voigt. C) Burgers. D.) A Generalised Maxwell (GM) model, where E_∞ is added to prevent the system acting as a maxwell fluid, as the maxwell arms would otherwise deform indefinitely as $t \rightarrow \infty$. A GM with a single maxwell arm is known as a simple linear solid or Zener model, and is the model used in this project.

Viscoelastic ECM-ECM interactions

A Simple Linear Solid model is employed in this work to determine the dynamics of ECM-ECM interactions. Compared to other common lumped parameter models, such as the Kelvin-Voigt or Burgers models, the Generalised Maxwell (GM) model has had increasingly dominant adoption in the hydrogel modelling literature [32]. These different model architectures are depicted in figure 2.5. GM models offer a unique advantage for hydrogel modelling applications, where each Maxwell arm can be tuned to the specific relaxation time, capturing different aspects of a viscoelastic material's dynamics. It has been theorised that the aqueous component of a hydrogel is encapsulated by shorter relaxation times, while the movement of large collagen fibres and other proteins captured by longer relaxation times [32]. Tirella et al. [33] used an experimental parameter fitting method (the epsilon dot method) to fit the parameters of a single arm (simple linear solid) and two arm GM model. Their results showed, for the two arm GM fitted to the liver tissue, the two relaxation rate were very similar. This was in-contrast to the gelatin sample which had distinct relaxation rates for each arm of the 2 arm GM. This shows that extending the GM with additional arms, can for certain materials, improve modelling accuracy. However, work by Fertela et al [7] has later found that on a smaller length scale, there is reduced poroelastic contribution and successfully fit a single term Prony series with minimal error compared to experimental results. Therefore in this work, to avoid the increase computational demand of higher order GM models, a GM with a single maxwell arm was used (known as a simple linear solid or zener model [34]). Discussion of this choice is continued in the further work section. [Reference section when available](#)

Consistently, the constitutive equation relevant in this case is

$$(2.23) \quad E(t) = E_0 + E_1 e^{-t/\tau}$$

where t is time under extension or compression, E_0 is the long-time (relaxed) modulus, representing the stiffness after all viscous dissipation. E_1 is the relaxation modulus, which is the viscous

component which decays on timescale τ . This timescale is defined for a Maxwell arm as $\tau = \eta_{VE}/E_1$, where η_{VE} is the constitutive dashpot viscosity. From these variables, additional physically relevant properties can be derived, for example, the instantaneous (unrelaxed) modulus is $E_u = E_0 + E_1$. Which is the full stiffness at $t = 0$, immediately after a deformation.

In this ABM, the ghost nodes are connected with a corresponding SLS model if they are within $1.6 \times h_g$, where h_g is the initial ghost node spacing. The parameters E_0 , E_1 and τ are taken from the AFM stress-relaxation measurements by Fertela et al. [7], from interpenetrating polymer network (IPN, starPEG–heparin) hydrogel microgels of diameter $67.8 \pm 2.4 \mu\text{m}$.

The discretised SLS equation consists of two rest lengths. s_0^{jk} which is the *permanent* rest length, from the initial ECM node spacing (h_g); and the *evolving dashpot length* $s_{jk}(t)$, which relaxes overtime toward the current separation of a given pair of nodes $\langle j, k \rangle$.

The internode spring force is then split into the two SLS arms,

$$(2.24) \quad \mathbf{F}_{jk}^{\text{ve}} = \left[E_0(d_{jk} - s_0^{jk}) + E_1(d_{jk} - s_{jk}(t)) \right] \Lambda_{jk}^{\text{gg}} \hat{\mathbf{r}}_{jk}$$

where $d_{jk} = \|\mathbf{x}_k - \mathbf{x}_j\|$ is the current pair distance and Λ_{jk}^{gg} is the anisotropic modulation factor

$$(2.25) \quad \Lambda_{jk}^{\text{gg}} = 1 - \alpha \bar{\alpha}_{jk} (1 - |\cos \theta|)$$

with β the anisotropy strength, $\bar{\alpha}_{jk}$ the mean fibre anisotropy of nodes j and k , and θ the angle between $\hat{\mathbf{r}}_{jk}$ and the mean fibre direction. At $t = 0$, $s_{jk} = s_0^{jk}$, so the force is $E_u(d_{jk} - s_0^{jk}) \Lambda_{jk}^{\text{gg}} \hat{\mathbf{r}}_{jk}$. As $t \rightarrow \infty$, $s_{jk} \rightarrow d_{jk}$, leaving the non-zero equilibrium force $E_0(d_{jk} - s_0^{jk}) \Lambda_{jk}^{\text{gg}} \hat{\mathbf{r}}_{jk}$, confirming solid-like behaviour and preventing the ECM population simply acting as a viscous fluid. To evaluate the force equation (2.24), the rest length s_{jk} must be found. This evolving dashpot length obeys

$$(2.26) \quad \frac{ds_{jk}}{dt} = \frac{d_{jk} - s_{jk}}{\tau}$$

As shown in appendix? The exact exponential integrator, which solves this linear ODE analytically over each interval is

$$(2.27) \quad s_{jk}^{n+1} = d_{jk}^n + (s_{jk}^n - d_{jk}^n) e^{-\Delta t/\tau}$$

Now that s_{jk} can be found at each time interval the instantaneous force in equation 2.24 is fully defined and is used with the inertia-free over-damped regime used in this project to determine the position updates of each ghost node. For each ECM ghost this update rule, combining both ECM-ECM and cell-ECM interaction forces, becomes

$$(2.28) \quad \eta_{\text{drag}} \frac{d\mathbf{x}_j^g}{dt} = \sum_{k \in \mathcal{G}(j)} \mathbf{F}_{jk}^{\text{ve}} + \sum_{i \in \mathcal{C}(j)} \mathbf{F}_{ji}^{\text{cg,react}}$$

where η_{drag} is the viscous drag coefficient controlling how quickly nodes reach mechanical equilibrium, and $\mathcal{C}(j)$ is the set of cells exerting reaction forces on ghost node j . Note that η_{drag} is an equation-of-motion parameter distinct from the constitutive dashpot viscosity $\eta_{VE} = E_1\tau$, the former governs the rate at which nodes move to equilibrium; the latter governs whether the material relaxes stress. Use of both drag coefficients is required to conform to the quasi-static regime used implemented in CHASTE thus far, however this is essentially applying the viscous drag twice, opening room for a improved modelling architecture.